

Venugopal K

+91 8074394544 | venu170305@gmail.com | linkedin.com/in/venugopal-k-0bba49287/ | github.com/Venugopal-2005

Final-year Computer Science student with hands-on experience in Java, Spring Boot, REST APIs, microservices, SQL, and Docker. Experienced in building secure backend applications with authentication, database integration, and containerized deployment.

EDUCATION

SRM University

Bachelor of Technology in Computer Science (CGPA: 8.45/10)

Chennai, TN

Aug 2023 – May 2027

EXPERIENCE

Computer Vision Intern

Dec 2025 – Jan 2026

MSPL Limited

Hosapete

- Developed an AI-powered Smart Employee Vehicle Safety & Access Control System using YOLO and OCR.
- Implemented real-time helmet detection and number plate recognition for industrial safety compliance.
- Contributed to AI-based automation solutions for enterprise applications using real-time object detection.

Python Web Development Intern

Aug 2025 – Oct 2025

Eduskills Academy – AICTE

- Developed web applications using Python and modern web development frameworks
- Strengthened backend development, database integration, and API implementation skills.
- Worked on real-world web development projects following industry best practices.

PROJECTS

Global Digital Banking System – Spring Boot Microservices

Jun 2026 – July 2026

Github: <https://github.com/Venugopal-2005/global-digital-banking-microservices-springboot>

Tech Stack: Java 17, Spring Boot, Spring Security, Spring JDBC, REST APIs, JWT, PostgreSQL, Docker, React

- Developed a distributed digital banking platform using **Spring Boot Microservices** with services for authentication, accounts, transactions, and API Gateway.
- Implemented JWT-based authentication and secure REST APIs using Spring Security.
- Built banking features including account management, fund transfers, deposits, and withdrawals with PostgreSQL integration.
- Containerized services using Docker Compose and managed database migrations with Flyway.
- Documented APIs using Swagger/OpenAPI and followed scalable microservices architecture.

Enterprise Distributed Job Scheduler

July 2026 – Present

Github: <https://github.com/Venugopal-2005/Distributed-Job-Scheduler#enterprise-distributed-job-scheduler>

Tech Stack: Java 21, Spring Boot 3, Spring Data JPA, PostgreSQL, Redis, Docker, Flyway, JWT

- Developed an enterprise-grade distributed job scheduler using Spring Boot with modular clean architecture and multi-tenant support.
- Implemented **JWT-based authentication**, Role-Based Access Control (RBAC), and RESTful APIs for secure job management.
- Built reliable job scheduling with retry mechanisms, dead-letter queue (DLQ), worker coordination, and heartbeat monitoring.
- Integrated PostgreSQL, Redis, Flyway, and Docker Compose for persistence, caching, database migrations, and containerized deployment; added monitoring and API documentation using Spring Boot Actuator, Swagger or OpenAPI.

Smart Employee Vehicle Safety & Access Control System

Dec 2025 – Jan 2026

GitHub: github.com/Venugopal-2005/Smart-Employee-Safety-Management

Tech Stack: Python, YOLO, OpenCV, OCR

- Developed an AI-powered vehicle safety and access control system using Computer Vision.
- Implemented real-time Helmet Detection and Number Plate Recognition using YOLO and OCR.
- Automated smart gate access based on safety compliance for industrial environments.

Smart Attendance Management System

May 2026 – Jun 2026

GitHub: <https://github.com/Venugopal-2005/Smart-Attendance-Management-System>

Tech Stack: React, FastAPI, OpenCV, Face Recognition

- Developed an AI-powered attendance management system using facial recognition.
- Automated attendance tracking through real-time face detection and identification.
- Reduced manual attendance effort by approximately 90%.

TECHNICAL SKILLS

- **Programming Languages**: Java, Python, C++, SQL
- **Backend & Frameworks**: Spring Boot, Spring Security, Spring JDBC, Spring Data JPA, REST APIs, JWT, Maven
- **Databases**: MySQL, PostgreSQL, SQLite
- **AI & Computer Vision**: OpenCV, YOLO, OCR, Face Recognition
- **DevOps & Tools**: Docker, Docker Compose, Git, GitHub, IntelliJ IDEA, VS Code, Postman, Flyway, Swagger, OpenAPI
- **CS Fundamentals**: Data Structures & Algorithms, OOP, DBMS, Computer Networks, Operating Systems
- **Other**: Agile Methodologies, Problem Solving

CERTIFICATIONS

- **NPTEL**: Introduction to Machine Learning
- **Scaler Topics** — DBMS Course – Master the Fundamentals and Advanced Concepts
- **Cisco Networking Academy** — Networking Basics
- **HP LIFE** — Agile Project Management
- **Fortinet Training Institute** — FortiGate 7.6 Operator Self-Paced
- **Udemy** — Beginning C++ Programming – From Beginner to Beyond

PARTICIPATIONS

- **Hacktrax Hackathon** – Alexa Developers SRM (24–25 March 2025)
- **Adobe India Hackathon** – Adobe (10 October 2024)
- **CreaTech 2024** – Larsen & Toubro (2024)
- **Pair Programming Competition** – Department of Computing Technologies, SRMIST (1–3 August 2024)